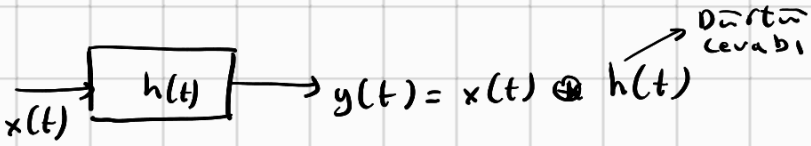
Lineer sistemlerden işaretlerin iletilmesi



$$Y(f) = X(f) \cdot H(f)$$

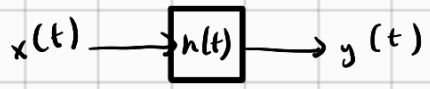
↳ Transfer fonk

$$|Y(f)|\, e^{j\theta_y(f)} = |X(f)|\, e^{j\theta_x(f)} \cdot |H(f)|\, e^{j\theta_h(f)}$$

$$|Y(f)| = |X(f)| \cdot |H(f)|$$

$$\theta_y(f) = \theta_x(f) + \theta_h(f)$$

Örnek:



$x(t) = 3 + 5\cos(3t + \frac{\pi}{4})$ $\qquad H(f) = j2\pi f = 10 \cdot j2\pi f \Rightarrow h(t) = 10\frac{d}{dt}(\,)$

a) $x(t)$'nin periyodu? $y(t)$'nin periyodu?

b) $P_x = ?$ $\qquad$ c) $y(t) = ?$ $\qquad$ d) $P_y = ?$

a) $x(t) = 3 + 5\cos(3t + \frac{\pi}{4})$ $\qquad \omega_0 = 3 = 2\pi f_0 = \frac{2\pi}{T_0} \longrightarrow T_{0x} = \frac{2\pi}{3}$

(ω_0)

* LZD sistemde çıkışın periyodu girişle aynıdır.

LZD $\Rightarrow T_{0,y} = T_{0,x} = \frac{2\pi}{3}$

b) $P_x = 3^2 + \frac{5^2}{2} = 21{,}5$ watt

$\omega_0 = 3 = 2\pi f_0 \Rightarrow f_0 = \frac{3}{2\pi}$

$$x(t) = \underbrace{3}_{x_0} + \underbrace{\frac{5}{2} e^{j\pi/4}}_{x_1} e^{j3t} + \underbrace{\frac{5}{2} e^{-j\pi/4}}_{x_2} e^{-j3t}$$

(e^{j3t}: ω_0, e^{-j3t}: $-\omega_0$)

$$\rightarrow P_x = \sum_n |x_n|^2 = 3^2 + \left(\frac{5}{2}\right)^2 + \left(\frac{5}{2}\right)^2 = 21{,}5 \text{ watt}$$

c) $Y(f) = X(f) \cdot H(f)$

$H(f) = j2\pi f$

$$X(f) = 3\delta(f) + \frac{5}{2} e^{j\pi/4} \delta(f - f_0) + \frac{5}{2} \cdot e^{-j\pi/4} \delta(f + f_0)$$

$$y(f) = H(f) \cdot X(f) = (j2\pi f)\left[3\delta(f) + \frac{5}{2} e^{j\pi/4} \delta(f - f_0) + \frac{5}{2} e^{-j\pi/4} \delta(f + f_0)\right]$$

$$= 3\delta(f) \cdot j2\pi(0) + j2\pi\left(\frac{3}{2\pi}\right) \cdot \frac{5}{2} e^{j\pi/4} \delta(f - f_0) + j2\pi\left(-\frac{3}{2\pi}\right) \frac{5}{2} e^{-j\pi/4} \delta(f + f_0)$$

($3\delta(f) \cdot j2\pi(0)$: 0)

$$= 75 j e^{j3\pi/4}\,\delta(f-f_0) - 75 j e^{-j\frac{\pi}{4}}\,\delta(f+f_0)$$

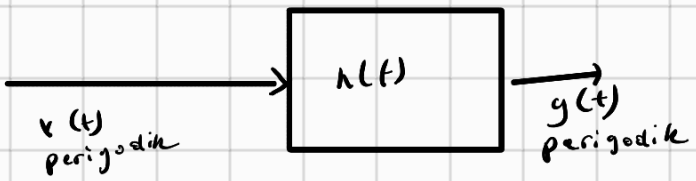
$e^{j3\pi/4}$, $e^{-j3\pi/4}$

$$Y(f) = 75 e^{j3\pi/4}\,\delta(f-f_0) + 75 e^{-j\frac{3\pi}{4}}\,\delta(f+f_0)$$

F^{-1} ↓

$$y(t) = 150\cos\left(2\pi f_0 t + \frac{3\pi}{4}\right)$$

$$P_y = \frac{150^2}{2}\ \text{watt} = 11250$$



$$x(t) = \sum_n x_n e^{j2\pi n f_0 t} \qquad y(t) = \sum_n y_n e^{j2\pi n f_0 t}$$

$$X(f) = \sum_n x_n \delta(f-nf_0) \qquad Y(f) = \sum_n y_n \delta(f-nf_0)$$

$$Y(f) = H(f)X(f) = H(f)\sum_n x_n \delta(f-nf_0)$$

$$= \sum_n x_n \underbrace{H(f)\,\delta(f-nf_0)}_{H(nf_0)\cdot\delta(f-nf_0)}$$

$$\boxed{Y(f) = \sum_n \underbrace{x_n H(nf_0)}_{y_n}\,\delta(f-nf_0)} \xrightarrow{F^{-1}} y(t) = \sum_n y_n e^{j2\pi n f_0 t}$$

Önceki ex 2.2.6 inceleyiniz (58-59 1. baskı)

Enerji ve Enerji yoğunluk fonksiyonu (EYF)

Energy Spektral Density (ESD)

$$E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} x(t)\cdot x^*(t)\,dt =$$

$$= \int_{-\infty}^{\infty} x(t)\left[\underbrace{\int_{-\infty}^{+\infty} X(f)\,e^{j2\pi f t}\,df}_{\int_{-\infty}^{\infty} X^*(f)\,e^{-j2\pi f t}}\right]^* dt$$

$$= \int_{-\infty}^{\infty} X^*(f) \underbrace{\int_{-\infty}^{\infty} x(t) e^{-j2\pi ft}\, dt}_{X(f)}\, df$$

$$= \int_{-\infty}^{\infty} |X(f)|^2\, df$$

Parseval t: $E_x = \int_{-\infty}^{\infty} |x(t)|^2\, dt = \int_{-\infty}^{\infty} \underbrace{|X(f)|^2}_{\mathcal{E}_x(f):\ EY\ F}\, df$

Enerji Tayf yoğunluğu F (EY F)

Enerji işaretleri için özilişki fonk (autocorelation f)

$R_x(\tau) = \int_{-\infty}^{+\infty} x(t) \cdot x^*(t-\tau)\, dt = \int_{-\infty}^{\infty} x(t)\, x^*(t+\tau)\, dt$

$R_x(\tau)$: $x(t)$'nin özilişki fonk

$= \int_{-\infty}^{+\infty} x^*(t)\, x(t+\tau)\, dt$

$R_x(-\tau) = R_x(\tau)$ çiftfonksiyonudur

$R_x(\tau) = x(\tau) \circledast x(-\tau)$

$x(t)$ reel olsun $\rightarrow$ $R_x(\tau) = x(\tau) \circledast x(-\tau)$

$\downarrow F$

$F\{R_x(\tau)\} = X(f) \cdot \frac{1}{|-1|} X\left(\frac{f}{-1}\right)$

$= X(f) \cdot \underbrace{X(-f)}_{X^*(f)}$

$x(t)$ reelse $X(f) = X^*(-f)$

$F^{-1}\{R_x(\tau)\} = |X(f)|^2 = \mathcal{E}_x(f)$

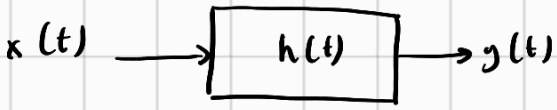
$$\boxed{R_x(\tau) \xleftrightarrow{F} \mathcal{E}_x(f) = |X(f)|^2}$$

$R_x(\tau) = \int_{-\infty}^{\infty} x(t)\, x^*(t-\tau)\, dt = \int_{-\infty}^{\infty} |x(t)|^2\, dt = E_x$

(τ yerine 0, t yerine 0 olsun)

Not: Enerji yoğunluk fonksiyonu max değeri $\tau = 0$'da olur

$$\boxed{E_x = R_x(0) = \int_{-\infty}^{+\infty} |x(t)|^2\, dt = \int_{-\infty}^{\infty} |X(f)|^2\, df = \int_{-\infty}^{+\infty} \mathcal{E}_x(f)\, df}$$



$y(t) = x(t) \circledast h(t)$

$Y(f) = X(f) \cdot H(f)$

$|Y(f)|^2 = |X(f)|^2 \, |H(f)|^2$

$\mathcal{E}_y(f) = \mathcal{E}_x(f) \, |H(f)|^2$

$\downarrow F^{-1}$

$F^{-1}\{\mathcal{E}_y(f)\} = F^{-1}\{\mathcal{E}_x(f)\} \circledast F^{-1}\{|H(f)^2|\}$

$R_y(z) \qquad R_x(z) \qquad R_h(z)$

$R_y(z) = y(z) \circledast y^A(-z) = R_x(z) \circledast R_h(z)$

Örnek

$x(t) = e^{-at}u(t), a>0$

a) $\mathcal{E}_x(f) = ?$ $E_x = ?$ b) $R_x(z) = ?$ $\mathcal{E}_x(f) = ?$ $E_x = ?$

a) $\mathcal{E}_x(f) = |X(f)|^2$ $\qquad X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft} dt$

$= \int_{-\infty}^{\infty} e^{-at} \underbrace{u(t)}_{\substack{=1 \; t\geq 0 \\ 0 \; t<0}} e^{-j2\pi ft} dt$

$= \int_0^{\infty} e^{-(a+j2\pi f)t} dt = \frac{e^{-(a+j2\pi f)t}}{-(a+j2\pi f)} \Big|_0^{\infty}$

weak distortion

$r = a+jb$

$|r| = \sqrt{a^2+b^2}$

$\theta_r = \arctan\left(\frac{b}{a}\right)$

$r = |r| e^{j\theta_r}$

$r^* = a - jb$

$= \frac{e^{-\infty} - e^{0}}{-(a+j2\pi f)} = \frac{1}{a+j2\pi f}$ ($e^{-\infty} = 0$)

$X(f) = \frac{1}{a+j2\pi f} \Rightarrow \mathcal{E}_x(f) = |X(f)|^2 = \frac{1}{a^2+(2\pi f)^2}$

$E_x = \int_{-\infty}^{\infty} \mathcal{E}_x(f) df = \int_{-\infty}^{\infty} \frac{df}{a^2+(2\pi)^2 f^2} = \frac{1}{2\pi a} \arctan\left(\frac{2\pi f}{a}\right) \Big|_{-\infty}^{\infty}$

$\int \frac{dx}{a^2+b^2x^2} = \frac{1}{ab} \arctan\left(\frac{bx}{a}\right)$

$= \frac{1}{2\pi a}\left[\frac{\pi}{2} - \left(-\frac{\pi}{2}\right)\right] = \boxed{\frac{1}{2a}}$

$E_x = \frac{1}{2a}$

Sağlama: $E_x = \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} e^{-2at} u(t)\, dt$

$= \frac{e^{-2at}}{-2a}\Big|_0^{\infty} = -\frac{1}{2a}\left(\underbrace{e^{-\infty}}_{0} - \underbrace{e^{-0}}_{1}\right)$

$= \frac{1}{2a}$

b) $R_x(z) = \int_{-\infty}^{\infty} x(t)\, x^*(t-z)\, dt = \int_{-\infty}^{\infty} e^{-at} u(t) \cdot e^{-a(t-z)} u(t-z)\, dt$

$u(t) = 1 \quad t \geq 0$

$u(t-z) = 1 \quad t - z \geq 0 \quad t \geq z$

$= e^{az} \int_{z}^{\infty} e^{-2at}\, dt =$

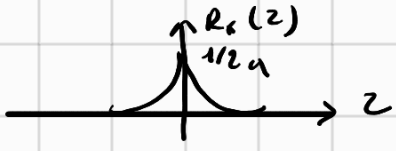
$z > 0$ pozitif için

$\rightarrow e^{az} \int_{z}^{\infty} e^{-2at}\, dt$

$= e^{az} \frac{e^{-2at}}{-2a}\Big|_z^{\infty} = -\frac{1}{2a} e^{az}\left[e^{-\infty} - e^{-2az}\right]$

$= \frac{e^{-az}}{2a}, \quad z > 0$

Çift fonksiyon olduğundan $R_x(z) = \frac{e^{-a|z|}}{2a} \quad \forall z$



$E_x = R_x(0) = 1/2a$

$R_x(z) = \frac{e^{-a|z|}}{2a} \xleftrightarrow{F} \mathcal{E}_x(f) = \frac{1}{a^2 + (2\pi f)^2}$

$e^{-a|z|}, z > 0 \xleftrightarrow{F} \frac{2a}{a^2 + (2\pi f)^2}$

Güç ve Güç yoğunluk fonk (GYF)

Power Spectral density (PSD)



$$x_T(t) = \begin{cases} x(t), & |t| \le \frac{T}{2} \\ 0, & |t| > \frac{T}{2} \end{cases}$$

$$x(t) \text{ reel} \Rightarrow P_x = \lim_{T\to\infty} \frac{1}{T} \underbrace{\int_{-T/2}^{T/2} x^2(t)\,dt}_{E_{xT}} = \lim_{T\to\infty} \frac{E_{xT}}{T}$$

$$x_T(t) \overset{F}{\longleftrightarrow} X_T(f)$$

$$E_{xT} = \int_{-\infty}^{\infty} x_T^2(t)\,dt = \int_{-\infty}^{\infty} |X_T(f)|^2\,df$$

$$P_x = \lim_{T\to\infty} \frac{E_{xT}}{T} = \lim_{T\to\infty} \frac{1}{T} \left[\int_{-\infty}^{\infty} |X_T(f)|^2\,df \right.$$

$T \nearrow \Rightarrow x_T(t)$'nin süresi $\nearrow \Rightarrow E_{xT} \nearrow \Rightarrow |X_T(f)|^2 \nearrow$

$$P_x = \int_{-\infty}^{+\infty} \underbrace{\lim_{T\to\infty} \frac{1}{T} |X_T(f)|^2}_{= S_x(f)}\,df = \int_{-\infty}^{+\infty} S_x(f)\,df$$

Güç yoğunluk fonk

EYF, GYF pozitif, reel, çift f.

x(t) birimi volt $\rightarrow$ Volt^2/Hz

Güç işaretlerinin Özilişki fonk

$x(t)$ güç işareti $\Rightarrow$

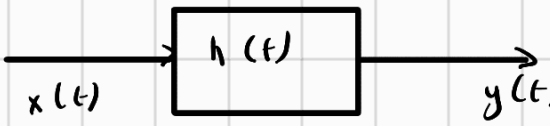
$$R_x(\tau) = \lim_{T\to\infty} \frac{1}{T}\int_{-T/2}^{T/2} x(t)\cdot x^*(t-\tau)\,dt$$

$$R_x(-\tau) = R_x(\tau)$$

$$R_x(\tau) \xleftrightarrow{F} S_x(f) = \lim_{T\to\infty} \frac{|X_T(f)|^2}{T}$$

$$R_x(0) = \lim_{T\to\infty} \frac{1}{T}\int_{-T/2}^{T/2} |x(t)|^2\,dt = P_x$$

$$P_x = \lim_{T\to\infty} \frac{1}{T}\int_{-T/2}^{T/2} |x(t)|^2\,dt = R_x(0) = \int_{-\infty}^{\infty} S_x(f)\,df$$



güç işareti:

güç işareti: $R_y(\tau) = \lim_{T\to\infty} \frac{1}{T}\int_{-T/2}^{T/2} y(t)\,y^*(t-\tau)\,dt$

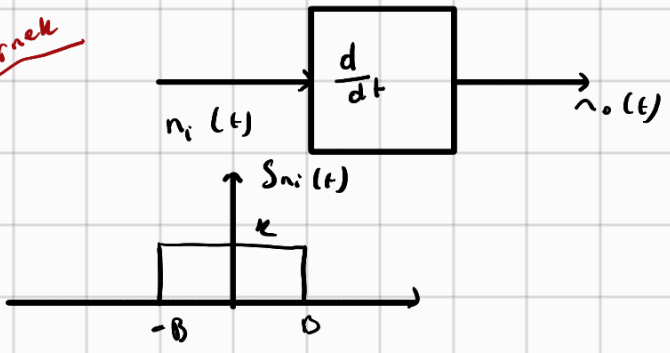
$$R_y(\tau) = R_x(\tau) \circledast \underbrace{h(\tau) \circledast h^*(-\tau)}_{R_h(\tau)}$$

F

$$S_y(f) = S_x(f)\cdot H(f)\cdot H^*(f)$$

$$\boxed{S_y(f) = S_x(f)\,|H(f)|^2}$$

Örnek



$$H(f) = j2\pi f$$

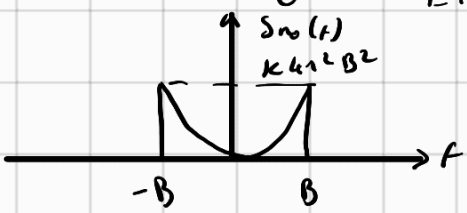
$n_i(t)$ is noise signal with PSD

$$S_{ni}(f) = \begin{cases} k & |f| \le B \\ 0 & \text{else} \end{cases}$$

$S_{no}(f) = ?$ $\quad P_{no} = ?$

$$S_{n_o}(f) = S_{ni}(f)\,|H(f)|^2$$

$$= \begin{cases} k\cdot 4\pi^2 f^2 & |f| \le B \\ 0 & \text{Else} \end{cases}$$



$$P_{no} = \int_{-\infty}^{\infty} S_{no}(f)\,df = \int_{-B}^{B} k4\pi^2B^2 f^2\,df$$

$$= \frac{8k\pi^2B^3}{3}$$

Örnek: $x(t)$ periyodik $\Rightarrow R_x(\tau) = ?$ $S_x(f) = ?$

$\Downarrow$ (yürür)

$$R_x(\tau) = \frac{1}{T_0}\int_{-T_0/2}^{T_0/2} x(t)\cdot x^*(t-\tau)\,dt$$

$x(t)$ periyodik $\Rightarrow$ $x(t) = \sum_{n=-\infty}^{\infty} x_n e^{j2\pi n f_0 t}$

$$R_x(\tau) = \frac{1}{T_0}\int_{-T_0/2}^{T_0/2} \underbrace{\sum_{n=-\infty}^{\infty} x_n\cdot e^{j2\pi n f_0 t}}_{x(t)} \cdot \underbrace{\sum_{m=-\infty}^{+\infty} x_m^* e^{-j2\pi m f_0(t-\tau)}}_{x^*(t)}\,dt$$

$$= \frac{1}{T_0}\sum_{n=-\infty}^{+\infty}\sum_{m=-\infty}^{\infty} x_n x_m^* e^{j2\pi m f_0\tau}\int_{-T_0/2}^{T_0/2} e^{j2\pi n f_0 t}\cdot e^{-j2\pi m f_0 t}\,dt$$

$$\int_{-T_0/2}^{T_0/2} e^{j2\pi(n-m)f_0 t}\,dt = \begin{cases} T_0 & n=m \\ 0 & n\neq m \end{cases}$$

$$\boxed{R_x(\tau) = \sum_{n=-\infty}^{+\infty} |x_n|^2 e^{j2\pi n f_0\tau}} \longleftrightarrow S_x(f) = \sum_{n=-\infty}^{+\infty} |x_n|^2\,\delta(f-nf_0)$$

Periyodik işaretin öz fonksiyonu periyodiktir

$$\rightarrow P_x = \int_{-\infty}^{+\infty} S_x(f)\,df = \int_{-\infty}^{\infty}\sum_{n=-\infty}^{\infty} |x_n|^2\,\delta(f-nf_0)\,df$$

$$= \sum_n |x_n|^2 \underbrace{\int_{-\infty}^{+\infty}\delta(f-nf_0)\,df}_{1} \qquad P_x = \sum_{n=-\infty}^{\infty} |x_n|^2$$